TRIGONOMETRY

(त्रिकोणमिति)

SHEET-03

CLASS ROOM SHEETS / FOR ALL EXAMS

UPDATED

TRIGONOMETRY (त्रिकोणमिति)

CLASSROOM SHEET-03

TYPE-I

1. Select the INCORRECT formula from the following options.

SSC CGL 17/07/2023 (Shift-01)

(a) $\sec^2\theta - \tan^2\theta = 1$
(b) $\sin^2\theta + \cos^2\theta = 1$
(c) $\text{cosec}^2\theta - \cot^2\theta = 1$
(d) $\sec^2\theta + \cos^2\theta = 1$

2. Which of the following is equal to $\frac{1}{\tan\theta} + \tan\theta$?

SSC CGL MAINS (08/08/2022

(a) $\frac{\cos ec\theta}{\sec\theta}$ (b) $\tan^2\theta$
(c) $\sec\theta \cdot \text{cosec}\,\theta$ (d) 1

3. Which of the following is equal to $\left[\frac{\cos\theta}{\sin\theta} + \frac{\sin\theta}{\cos\theta}\right]$?

SSC CGL MAINS (08/08/2022)

(a) $\cot\theta \cdot \sec\theta$ (b) $\text{cosec}\,\theta \cdot \sec\theta$
(c) $\sec\theta \cdot \tan\theta$ (d) $\text{cosec}\,\theta \cdot \tan\theta$

4. Which of the following is equal to secA – cosA?

SSC CGL MAINS (08/08/2022)

(a) sinA. cotA (b) cotA. cosA
(c) tanA. sinA (d) cosA sinA

5. $[(\sin\theta \tan\theta + \cos\theta)^2 - 1]$ is equal to:

SSC Phase X 02/08/2022 (Shift- 02)

(a) $\sec^2\theta$ (b) $\sec\theta$
(c) $\text{cosec}\,\theta$ (d) $\tan^2\theta$

6. For $0° < \theta < 90°$, $\frac{1}{\cos\theta} + \frac{1}{\tan\theta - \sec\theta}$ is equal to:

SSC CGL 17/08/2021 (Shift- 02)

(a) $-\sec\theta$ (b) $\tan\theta$
(c) $\sec\theta$ (d) $-\tan\theta$

7. The value of $\left(\frac{1}{\sin\theta} + \frac{1}{\tan\theta}\right)\left(\frac{1}{\sin\theta} - \frac{1}{\tan\theta}\right)$ is:

SSC CGL 18/07/2023 (Shift-02)

(a) 0 (b) 2
(c) 3 (d) 1

8. (cosecA – cotA) (1+cosA) =?

SSC CGL 23/08/2021 (Shift- 02)

(a) cosecA (b) cosA
(c) sinA (d) cotA

9. $(\text{cosec}\theta - \cot\theta)^2 =$, $0° < \theta < 90°$

SSC CHSL 09/06/2022 (Shift- 03)

(a) $\frac{1-\sin\theta}{1+\cos\theta}$ (b) $\frac{1-\cos\theta}{1+\cos\theta}$
(c) $\frac{1+\cos\theta}{1-\cos\theta}$ (d) $\frac{1+\cos\theta}{1-\sin\theta}$

10. The given expression is equal to:

$\frac{(1+\tan^2 A)}{\cos ec^2 A.\tan A}$

SSC CHSL 03/08/2023 (Shift-03)

(a) $\sec^2 A$ (b) $\sec A$
(c) $\tan A$ (d) $\tan A$

TYPE-II

11. Find the value of the following expression.

$\sqrt{\frac{1+\sin\theta}{1-\sin\theta}}$

SSC CGL 05/12/2022 (Shift-04)

(a) $\sec\theta + \tan\theta$ (b) $\text{cosec}\theta + \tan\theta$
(c) $\text{cosec}\theta + \cot\theta$ (d) $\sec\theta + \cot\theta$

12. Find the value of $\sqrt{\frac{1-\sin 3\theta}{1+\sin 3\theta}}$

SSC CGL 21/07/2023 (Shift-02)

(a) $\sec 3\theta - \tan 3\theta$ (b) $(\sec 3\theta - \tan 3\theta)^2$
(c) $(\sec 3\theta - \tan 3\theta)^3$ (d) $\sec 3\theta + \tan 3\theta$

13. What is the value of

$\sqrt{\frac{\text{cosec}A+1}{\text{cosec}A-1}} + \sqrt{\frac{\text{cosec}A-1}{\text{cosec}A+1}} = ?$

SSC CHSL 13/03/2023 (Shift-01)

(a) 2 cosA (b) secA
(c) cosA (d) 2 secA

14. Simplify the given expression

$\sqrt{\frac{1+\cos P}{1-\cos P}}$

SSC CGL 25/07/2023 (Shift-03)
SSC CGL 03/12/2022 (Shift-04)
SSC CHSL 17/08/2023 (Shift-03)

(a) cosecP–cotP (b) secP–tanP
(c) secP+tanP (d) cosecP+cotP

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15. If A is an acute angle, then $\sqrt{\frac{1-\cos A}{1+\cos A}}+\sqrt{\frac{1+\cos A}{1-\cos A}}$ is equal to:

SSC CGL 05/12/2022 (Shift-02)
SSC CGL 19/09/2024 (Shift-03)

(a) 2cosecA (b) 2cosA
(c) 2secA (d) 2sinA

16. The given expression is equivalent to:

$$\sqrt{\frac{\sec\theta+1}{\sec\theta-1}}+\sqrt{\frac{\sec\theta-1}{\sec\theta+1}}$$

SSC CGL 18/09/2024 (Shift-03)

(a) 2sinθ (b) 2cosecθ
(c) 2tanθ (d) 2tanθ secθ

17. What is the value of $\frac{\tan A-\sin A}{\tan A+\sin A}$?

SSC CHSL 09/03/2023 (Shift-02)

(a) 1 (b) $\frac{\sec A+1}{\sec A-1}$
(c) $\frac{\sec A-1}{\sec A+1}$ (d) secA

18. Which of the following is equal to $\left[\frac{\tan\theta+\sec\theta-1}{\tan\theta-\sec\theta+1}\right]$?

SSC CGL TIER-II 08/08/2022

(a) $\frac{1+\cos\theta}{\sin\theta}$ (b) $\frac{1+\cot\theta}{\tan\theta}$
(c) $\frac{1+\sin\theta}{\cos\theta}$ (d) $\frac{1+\tan\theta}{\cot\theta}$

19. What is the value of $\frac{\tan\theta-\sec\theta+1}{\tan\theta+\sec\theta-1}$?

SSC CHSL 21/03/2023 (Shift-03)

(a) secθ + tanθ (b) $\frac{(1+\sin\theta)}{\cos\theta}$
(c) 2secθ (d) $\frac{\cos\theta}{(1+\sin\theta)}$

20. What is the value of $\frac{\cot\theta+\operatorname{cosec}\theta-1}{\cot\theta-\operatorname{cosec}\theta+1}$?

SSC CHSL 20/03/2023 (Shift-02)

(a) 2 secθ (b) 2 cosecθ
(c) 2 cotθ (d) (cosecθ + cotθ)

21. The value of $\frac{\sin\theta+\cos\theta-1}{\sin\theta-\cos\theta+1}\times\sqrt{\frac{1+\sin\theta}{1-\sin\theta}}$

SSC CGL 2019 Tier-II (16/10/2020)

(a) 1 (b) – 1
(c) – 2 (d) 2

22. Find the value of $\frac{\sin\theta+\cos\theta-1}{\sin\theta-\cos\theta+1}\times\frac{\tan^2\theta(\operatorname{cosec}^2\theta-1)}{\sec\theta-\tan\theta}$.

(a) 0 (b) – 1
(c) 1 (d) $\frac{1}{2}$

23. $\frac{\sin A-\cos A+1}{\sin A+\cos A-1}$=?

(a) secA + tanA (b) $\frac{1+\sin A}{\cos A}$
(c) $\frac{\sin\frac{A}{2}+\cos\frac{A}{2}}{\cos\frac{A}{2}-\sin\frac{A}{2}}$ (d) All of these

24. If $\frac{\sin\theta-\cos\theta+1}{\sin\theta+\cos\theta-1}$ = p secθ + q tanθ, where $0<\theta<\frac{\pi}{2}$, then what is p + q equal to?

CDS I 21/04/2024

(a) 0 (b) 1
(c) 2 (d) 4

25. $\frac{\sin2A-\cos2A+1}{\sin2A+\cos2A-1}$

(a) $\sin\frac{A}{2}$ (b) $\cos\frac{A}{2}$
(c) $\frac{\sin A+\cos A}{\cos A-\sin A}$ (d) $\frac{\sin A+\cos A}{\sin A-\cos A}$

26. $\frac{\sin A+\cos A-1}{\sin A-\cos A+1}$=?

(a) secA – tanA (b) $\frac{1-\sin A}{\cos A}$
(c) $\frac{\cos A}{1+\sin A}$ (d) All of these

27. $\frac{\cos A-\sin A+1}{\cos A+\sin A-1}$=?

(a) $\cot\frac{A}{2}$ (b) $\frac{1+\cos A}{\sin A}$
(c) cosec + cotA (d) All of the above

28. What is $\left(\frac{\cos\theta-\sin\theta+1}{\cos\theta+\sin\theta-1}\right)(\cot\theta-\operatorname{cosec}\theta)$ equal to?

CDS II 01/09/2024

(a) –1 (b) 0
(c) 1 (d) 2

29. $\frac{\cos A+\sin A-1}{\cos A-\sin A+1}$=?

(a) $\tan\frac{A}{2}$ (b) $\frac{1-\cos A}{\sin A}$
(c) cosec – cotA (d) All of these

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30. Simplify $\frac{1+\sin\theta-\cos\theta}{1+\sin\theta+\cos\theta}$.

SSC CGL 24/09/2024 (Shift-03)

(a) $\cos\frac{\theta}{2}$ (b) $\sin\frac{\theta}{2}$

(c) $\cot\frac{\theta}{2}$ (d) $\tan\frac{\theta}{2}$

31. What is the value of $\tan\frac{\theta}{2}$?

SSC CGL 26/07/2023 (Shift-04)

(a) $\frac{\cos\theta}{1-\sin\theta}$ (b) $\frac{\sin\theta}{1-\cos\theta}$

(c) $\frac{\cos\theta}{1-\cos\theta}$ (d) $\frac{\sin\theta}{1+\cos\theta}$

32. If $\tan\frac{A}{2} = x$, then find x.

SSC CGL 18/07/2023 (Shift-01)

(a) $\frac{\sqrt{1+\cos A}}{\sqrt{1-\cos A}}$ (b) $\frac{\sqrt{1-\sin A}}{\sqrt{1+\cos A}}$

(c) $\frac{\sqrt{1-\cos A}}{\sqrt{1+\cos A}}$ (d) $\frac{\sqrt{\cos A-1}}{\sqrt{1+\cos A}}$

33. The value of $1+\sqrt{\frac{\cot\theta+\cos\theta}{\cot\theta-\cos\theta}}$, if $0° < \theta < 90°$, is equal to:

SSC CGL 12/04/2022 (Shift-02)

(a) $1 - \sec\theta + \tan\theta$
(b) $1 - \sec\theta - \tan\theta$
(c) $1 + \sec\theta - \tan\theta$
(d) $1 + \sec\theta + \tan\theta$

34. Simplify: $\frac{\cos A}{1+\tan A} - \frac{\sin A}{1+\cot A}$

SSC CGL 03/12/2022 (Shift-01)

(a) tanA (b) cosA – sinA
(c) cosA. sinA (d) cosA + sinA

35. Simplify: $\frac{\cos A}{1-\tan A} + \frac{\sin A}{1-\cot A}$

SSC CGL 13/12/2022 (Shift-02)

(a) tanA – cotA (b) tanA + cotA
(c) sinA – cosA (d) sinA + cosA

36. Simplify the following expression:

$$\frac{\cos A}{1-\tan A} + \frac{\sin A}{1-\cot A} - \sin A$$

SSC CGL 19/04/2022 (Shift-03)

(a) 1 + sinA (b) (1 + sinA) cosA
(c) cosA (d) 1 + cosA

37. Solve the following:
(tanθ + secθ + 1)(cotθ – cosecθ + 1) =?

SSC CPO 29/06/2024 (Shift-01)

(a) 2 (b) 1
(c) 0 (d) –1

38. What is the value of
(1 + tanθ – secθ)(1 + cotθ + cosecθ)?

SSC CHSL 09/03/2023 (Shift-03)

(a) – 1 (b) 0
(c) 1 (d) 2

39. The value of
(1 + cot A – cosec A)(1 + tanA + sec A)–1 is:

SSC CPO 09/11/2022 (Shift-02)

(a) 1 (b) 2
(c) 3 (d) 0

40. The value of
(cosecA + cotA+1)(cosecA–cotA + 1) – 2cosecA is

SSC CGL 09/03/2020 (Shift-02)

(a) 4cosecA (b) 2
(c) 2cosecA (d) 0

41. Find the value of
(1 + cotA – cosecA) (1 + tanA + secA) –3($\sin^2$A + $\cos^2$A).

SSC CPO 27/06/2024 (Shift-02)

(a) –1 (b) 1
(c) 2 (d) –2

42. If A is an acute angle, which of the following is equal to $\frac{\sin A}{1+\cos A}$?

SSC CGL 09/09/2024 (Shift-03)

(a) $\frac{1-\cos A}{\sin A}$ (b) $\frac{1+\cos A}{\sin A}$

(c) $\frac{1-\sin A}{\cos A}$ (d) $\frac{1+\sin A}{\cos A}$

43. If $p = \frac{\sin A}{1+\cos A}$ then $\frac{\sin A}{1-\cos A}$ is equal to:

SSC CGL 26/09/2024 (Shift-02)

(a) $\frac{1}{p}$ (b) $\frac{1}{1-p}$

(c) $\frac{1}{p+1}$ (d) p – 1

44. Simplify: $\frac{\sin A}{1+\cos A}$ + cotA, where A is an acute angle.

SSC CGL 18/09/2024 (Shift-02)

(a) SinA (b) CosA
(c) sinA + cosA (d) cosecA

45. Find the value of $\sqrt{\frac{1-\tan A}{1+\tan A}}$

SSC CPO 04/10/2023 (Shift-02)

(a) $\sqrt{\frac{1+\sin A}{\cos A}}$ (b) $\sqrt{\frac{1+\sin 2A}{\cos 2A}}$

(c) $\sqrt{\frac{1-\sin A}{\cos A}}$ (d) $\sqrt{\frac{1-\sin 2A}{\cos 2A}}$

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TYPE - III

46. If $\tan\theta + \cot\theta = 2$ and θ is an acute angle, then the value of θ is:

SSC CGL 19/09/2024 (Shift-03)

(a) 30° (b) 15°
(c) 60° (d) 45°

47. If A is an acute angle and $\tan A + \cot A = 2$, find the value of $7\tan^3 A - 6\cot^3 A + 8\sec^2 A$.

SSC CGL 26/09/2024 (Shift-01)

(a) 6 (b) 17
(c) 16 (d) 7

48. If θ is an angle and $\sin\theta + \operatorname{cosec}\theta = 2$, then the value of $\sin^5\theta + \operatorname{cosec}^5\theta$ is:

SSC CPO 03/10/2023 (Shift-01)

(a) 2 (b) 10
(c) 4 (d) 5

49. If $\cos\theta + \sec\theta = 2$, then the value of $\sin^6\theta + \cos^6\theta$ is:

SSC CPO 10/11/2022 (Shift-03)

(a) $\frac{1}{3}$ (b) 0
(c) 1 (d) $\frac{1}{2}$

50. If $\tan\theta + \cot\theta = 6$, then the value of $\tan^2\theta + \cot^2\theta$ is ____ .

SSC CPO 28/06/2024 (Shift-02)

(a) 24 (b) 34
(c) 20 (d) 22

51. If $\sec\theta + \cos\theta = 32$, then $\sec^2\theta + \cos^2\theta$ is ____.

SSC CGL 14/07/2023 (Shift-04)

(a) 1000 (b) 1022
(c) 1024 (d) 1020

52. If $\sin\theta + \operatorname{cosec}\theta = \sqrt{5}$, then the value of $\sin^3\theta + \operatorname{cosec}^3\theta$ is:

SSC Phase-XII 26/06/2024 (Shift-03)

(a) $\sqrt{5}$ (b) 0
(c) $\frac{1}{\sqrt{5}}$ (d) $2\sqrt{5}$

53. If $\sin\theta - \operatorname{cosec}\theta = 6$, then what is the value of $\sin\theta + \operatorname{cosec}\theta$?

SSC CHSL 14/03/2023 (Shift-02)

(a) 6 (b) $\sqrt{40}$
(c) $\sqrt{34}$ (d) $\sqrt{38}$

54. If $3x = \sec A$ and $\frac{3}{x} = \tan A$ then $9\left(x^2 - \frac{1}{x^2}\right)$ is ____.

SSC CGL 27/07/2023 (Shift-02)

(a) 3 (b) 9
(c) 1 (d) $\frac{1}{9}$

TYPE - IV

55. If $\sec\theta + \tan\theta = 5$, then what is the value of $\sec\theta - \tan\theta$?

SSC CHSL 09/03/2023 (Shift-01)

(a) $\frac{1}{5}$ (b) $\frac{1}{3}$
(c) $\frac{1}{6}$ (d) $\frac{1}{2}$

56. If the value of $\operatorname{cosec} A + \cot A = m$, then the value of $\operatorname{cosec} A - \cot A$ is:

SSC CPO 10/11/2022 (Shift-02)

(a) $\frac{1}{m}$ (b) m
(c) $\sqrt{m}$ (d) m^2

57. If $\sec\theta + \tan\theta = x$, then find $\sin\theta$.

SSC CGL 25/09/2024 (Shift-03)

(a) $\frac{x^2+1}{1-x^2}$ (b) $\frac{x^2-1}{1+2x^2}$
(c) $\frac{x^2-1}{1+x^2}$ (d) $\frac{1-x^2}{1+x^2}$

58. If $\sec\theta + \tan\theta = 5$, $(\theta \neq 0)$ then $\sec\theta$ is equal to:

SSC CGL 14/07/2023 (Shift-04)

(a) $\left(5 + \frac{1}{5}\right)$ (b) $\frac{1}{2}\left(3 + \frac{1}{3}\right)$
(c) $\frac{1}{2}\left(5 + \frac{1}{5}\right)$ (d) $\left(3 + \frac{1}{3}\right)$

59. If $\sec\theta + \tan\theta = \sqrt{3}$ then find the positive value of $\sin\theta + \cos\theta$.

SSC CHSL 04/08/2023 (Shift-03)

(a) $\frac{\sqrt{3}}{2}$ (b) $\frac{2}{\sqrt{3}}$
(c) $\frac{2}{(1+\sqrt{3})}$ (d) $\frac{(1+\sqrt{3})}{2}$

60. If $\sec^2\theta + \tan^2\theta = \frac{25}{18}$, then the value of $\sec^4\theta - \tan^4\theta$ is:

SSC CGL 14/07/2023 (Shift-03)

(a) $\frac{18}{25}$ (b) $\frac{25}{12}$
(c) $\frac{25}{9}$ (d) $\frac{25}{18}$

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61. If $\text{cosec}\theta + \cot\theta = \frac{3}{2}$, then what is the value of $\text{cosec}\,\theta$?

SSC CGL 24/07/2023 (Shift-03)

(a) $\frac{13}{12}$ (b) $\frac{3}{2}$

(c) $\frac{11}{12}$ (d) $\frac{9}{13}$

62. If $\text{cosec}\theta + \cot\theta = m$, find the value of $\frac{m^2-1}{m^2+1}$.

SSC CGL 24/07/2023 (Shift-02)

(a) 1.1 (b) 2.0

(c) $\cos\theta$ (d) $-\sin\theta$

63. If $\text{cosec}A + \cot A = a\sqrt{b}$, then find the value of $\frac{(a^2b-1)}{(a^2b+1)}$.

SSC CHSL 17/08/2023 (Shift-02)

(a) cosA (b) tanA

(c) $\frac{1}{\sin A}$ (d) $\frac{1}{\cot A}$

64. If $\sec x + \tan x = 5$ and $\text{cosec}\,y - \cot y = \frac{1}{3}$, then find the value of $(\sec x + \text{cosec}\,y) - (\tan x - \cot y)$.

SSC CHSL 15/03/2023 (Shift-01)

(a) 2.2 (b) 3.2

(c) 4.2 (d) 3.1

TYPE - V

65. If $10\sin^2\theta + 6\cos^2\theta = 7$, $0 < \theta < 90°$, then find the value of $\tan\theta$.

SSC CGL 24/09/2024 (Shift-03)

(a) $\sqrt{3}$ (b) 1

(c) $\frac{1}{\sqrt{3}}$ (d) $\frac{1}{2}$

66. If $4\sin^2\theta = 3(1 + \cos\theta)$, $0° < \theta < 90°$, then what is the value of $\sqrt{15}\tan\theta + \frac{4}{\sqrt{15}}\sin\theta + 2\sec\theta$?

SSC CPO 28/06/2024 (Shift-02)

(a) 24 (b) $4\sqrt{15}$

(c) $\frac{8}{\sqrt{15}}$ (d) 8

67. If $7\cos^2\theta + 5\sin^2\theta - 6 = 0$, $(0° < \theta < 90°)$, then what is the value of $1+\sqrt{\frac{\sec\theta+\tan\theta}{\sec\theta-\tan\theta}}$?

SSC CGL 10/09/2024 (Shift-03)

(a) $\sqrt{2}+2$ (b) $\sqrt{2}+1$

(c) $\sqrt{2}-1$ (d) $\sqrt{2}-2$

68. If $\frac{\sin^2\theta - 3\sin\theta + 2}{\cos^2\theta} = 1$, where $0° < \theta < 90°$, then what is the value of $(\cos 2\theta - \sin 3\theta + \text{cosec}2\theta)$

SSC CGL 21/04/2022 (Shift-02)

(a) $\frac{3+4\sqrt{3}}{6}$ (b) $\frac{-3-4\sqrt{3}}{6}$

(c) $\frac{3-4\sqrt{3}}{6}$ (d) $\frac{-3+4\sqrt{3}}{6}$

69. If θ is an acute angle and $\sin\theta\cos\theta = 2\cos^3\theta - \frac{1}{4}\cos\theta$, then the value of $\sin\theta$ is:

SSC CHSL 10/08/2023 (Shift-02)

(a) $\frac{\sqrt{15}-1}{8}$ (b) $\frac{\sqrt{15}-1}{4}$

(c) $\frac{\sqrt{15}+1}{4}$ (d) $\frac{\sqrt{15}-1}{2}$

70. Find the value of θ, if $\sec^2\theta + (1-\sqrt{3})\tan\theta - (1+\sqrt{3}) = 0$, where θ is an acute angle.

SSC CHSL 05/08/2021 (Shift-03)

(a) 30° (b) 15°

(c) 45° (d) 60°

71. If $\frac{\sin^2\theta}{\cos^2\theta - 3\cos\theta + 2} = 1$, θ lies in the first quadrant, then the value of $\frac{\tan^2\frac{\theta}{2} + \sin^2\frac{\theta}{2}}{\tan\theta + \sin\theta}$ is:

SSC CGL TIER-II 03/02/2022

(a) $\frac{2\sqrt{3}}{27}$ (b) $\frac{5\sqrt{3}}{27}$

(c) $\frac{2\sqrt{3}}{9}$ (d) $\frac{7\sqrt{3}}{54}$

TYPE - VI

72. If $\sin A + \sin^2 A = 1$, then the value of $\cos^4 A + \cos^6 A$ is:

SSC CPO 04/10/2023 (Shift-01)

(a) 1 (b) cos A

(c) sin A (d) 0

73. If $\cos A + \cos^2 A = 1$, then the value of $\sin^4 A + \sin^6 A$ is:

SSC CGL 26/07/2023 (Shift-01)

(a) 2 (b) cosA

(c) 1 (d) sinA

74. If $\cot^4\theta - \cot^2\theta = 1$, then the value of $\cos^4\theta + \cos^2\theta$ is:

SSC CGL 12/09/2024 (Shift-02)

(a) 2 (b) 1

(c) $\frac{1}{\sqrt{3}}$ (d) $\frac{3}{2}$

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75. If $\cot^4A + \cot^2A = 1$ then find the value of $\cos^4A - 3\cos^2A$.

SSC CHSL TIER-II 02/11/2023

(a) 2 (b) −1
(c) 1 (d) −2

76. If $\sin^4x + \sin^2x = 1$, then what is the value of $\cot^4x + \cot^2x$?

SSC CHSL 17/03/2023 (Shift-02)

(a) − 2 (b) 2
(c) − 1 (d) 1

77. If $\tan^4\theta + \tan^2\theta = 1$, what is the value of $11(\cos^4\theta + \cos^2\theta)$?

SSC CGL 27/07/2023 (Shift-03)

(a) −11 (b) 8
(c) 0 (d) 11

78. If $\sin x + \sin^2x = 1$, then what is the value of $\cos^8x + 2\cos^6x + \cos^4x$?

SSC CHSL TIER-II 26/06/2023

(a) 1 (b) 0
(c) 4 (d) 2

79. If $\cos A = \sin^2A$, and $a\sin^{12}A + b\sin^{10}A + c\sin^8A + \sin^6A = 1$, then $a + b + c = ?$

SSC CPO 11/11/2022 (Shift-01)

(a) 7 (b) 8
(c) 9 (d) 6

TYPE - VII

80. If $\sin\theta - \cos\theta = \frac{1}{5}$, then find the value of $\sin\theta + \cos\theta$.

SSC CHSL 02/08/2023 (Shift-04)

(a) $\frac{5}{7}$ (b) $\frac{7}{5}$
(c) $\frac{5}{3}$ (d) $\frac{3}{5}$

81. If $\cos\theta - \sin\theta = \sqrt{2}\sin\theta$, then $(\cos\theta + \sin\theta)$ is:

SSC CPO 03/10/2023 (Shift-01)

(a) $-\sqrt{2}\sin\theta$ (b) $\sqrt{2}\tan\theta$
(c) $\sqrt{2}\cos\theta$ (d) $-\sqrt{2}\cos\theta$

82. If $4\cos\theta + 3\sin\theta = x$ and $4\sin\theta - 3\cos\theta = y$, find the value of $x^2 + y^2$.

SSC CPO 11/11/2022 (Shift-01)

(a) 16 (b) 9
(c) 25 (d) 1

83. If $\sin x + \cos x = \sqrt{2}\sin x$, then the value of $\sin x - \cos x$ is:

SSC CPO 28/06/2024 (Shift-02)

(a) $\sin x$ (b) $2\sin x$
(c) $-\sqrt{2}\cos x$ (d) $\sqrt{2}\cos x$

84. If $\sin\theta + \cos\theta = \frac{\sqrt{11}}{3}$, then the value of $(\cos\theta - \sin\theta)$ is:

SSC CPO 03/10/2023 (Shift-03)

(a) $\frac{\sqrt{5}}{3}$
(b) $\frac{7}{3}$
(c) $\frac{5}{3}$
(d) $\frac{\sqrt{7}}{3}$

85. If $a\cot\theta + b\operatorname{cosec}\theta = p$ and $b\cot\theta + a\operatorname{cosec}\theta = q$ then $p^2 - q^2$ is equal to _____.

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(a) $b^2 - a^2$
(b) $a^2 - b^2$
(c) $b - a$
(d) $a^2 + b^2$

ANSWER KEY

1.	(d)	2.	(c)	3.	(b)	4.	(c)	5.	(d)	6.	(d)	7.	(a)	8.	(c)	9.	(b)	10.	(c)
11.	(a)	12.	(a)	13.	(d)	14.	(d)	15.	(a)	16.	(b)	17.	(c)	18.	(c)	19.	(d)	20.	(d)
21.	(a)	22.	(c)	23.	(d)	24.	(c)	25.	(c)	26.	(d)	27.	(d)	28.	(a)	29.	(c)	30.	(d)
31.	(d)	32.	(c)	33.	(d)	34.	(b)	35.	(d)	36.	(c)	37.	(a)	38.	(d)	39.	(a)	40.	(b)
41.	(a)	42.	(a)	43.	(a)	44.	(d)	45.	(d)	46.	(d)	47.	(b)	48.	(a)	49.	(c)	50.	(b)
51.	(b)	52.	(d)	53.	(b)	54.	(c)	55.	(a)	56.	(a)	57.	(c)	58.	(c)	59.	(d)	60.	(d)
61.	(a)	62.	(c)	63.	(a)	64.	(b)	65.	(c)	66.	(a)	67.	(a)	68.	(d)	69.	(b)	70.	(d)
71.	(d)	72.	(c)	73.	(b)	74.	(b)	75.	(b)	76.	(d)	77.	(d)	78.	(a)	79.	(a)	80.	(b)
81.	(c)	82.	(c)	83.	(d)	84.	(d)	85.	(a)										

Maths by Aditya Ranjan sir

CALCULATION TECHNIQUES

TECHNIQUE – 01

♦ Cube Root of Non-perfect cube ♦

$$\sqrt[3]{N} = A \pm \frac{B}{3.A^2}$$

First find a number which is a perfect cube near to the number N, let it be A and the difference between N and A^3 is B.

Note: '+' sign is used if $N > A^3$ & '–' sign is used if $N < A^3$

Ex. $\sqrt[3]{60} = 4 - \frac{4}{3\times4^2} = \frac{188}{48} = 3.92$

Ex. $\sqrt[3]{72} = 4 + \frac{8}{3\times4^2} = \frac{200}{48} = 4.17$

♦ Do it Yourself ♦

(i) $\sqrt[3]{10}$ = ...

(ii) $\sqrt[3]{66}$ = ...

(iii) $\sqrt[3]{118}$ = ...

(iv) $\sqrt[3]{226}$ = ...

(v) $\sqrt[3]{525}$ = ...

(vi) $\sqrt[3]{1365}$ = ...

(vii) $\sqrt[3]{1500}$ = ...

TECHNIQUE – 02

♦ Square of two digit number ♦

+ Carry + Carry

$(ab)^2 = a^2/2ab/b^2$

1 digit, 1 digit

- **Square of 14**

Here, a = 1 & b = 4

$14^2 = 1^2/2.1.4/4^2 = 1/8/16 = 196$

- **Square of 23**

Here, a = 2 & b = 3

$23^2 = 2^2/2.2.3/3^2 = 4/12/9 = 529$

♦ Do it Yourself ♦

(i) 18^2 = ... (ii) 23^2 = ...

(iii) 27^2 = ... (iv) 34^2 = ...

(v) 38^2 = ... (vi) 43^2 = ...

(vii) 57^2 = ... (viii) 64^2 = ...

(ix) 66^2 = ... (x) 75^2 = ...

(xi) 78^2 = ... (xii) 84^2 = ...

SELF MENTORSHIP TABLE

	STEP-01	STEP-02	STEP-03	STEP-04	STEP-05	STEP-06
TOPIC NAME	VIDEOS	CLASSROOM SHEET	SSC MATHS	NOTES / FORMULA BOOK REVISION	CHAPTER TEST	PRO LEVEL PRACTICE SHEET
TRIGONOMETRY SHEET - 03						
TRIGONOMETRIC IDENTITIES (BASIC QUESTIONS)						
SOME SPECIAL CONCEPTS						
BASED ON ALGEBRAIC FORMULAE						
If $\sec\theta + \tan\theta = K$, Then $\sec\theta - \tan\theta = 1/K$, If $\operatorname{cosec}\theta - \cot\theta = K$, Then $\operatorname{cosec}\theta + \cot\theta = 1/K$						
BASED ON QUADRATIC EQUATIONS						
BAHUBALI CONCEPT						
$a\sin\theta + b\cos\theta = x$, $a\cos\theta - b\sin\theta = y$, Then, $a^2 + b^2 = x^2 + y^2$						